Aejandro De La Torre
LIS 440
wk14: EXERCISE -- Pandas Aggregating -- Summarizing and Grouping
12/3/25

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path

# Print version of pandas
print("Alejandro De La Torre's code....")
print(pd.__version__)


# ----------------------------------------
# File & Path Setup
# ----------------------------------------

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'
output = root / 'output'
code = root / 'code'


# ----------------------------------------
# Analysis: Totals and Groupings
# ----------------------------------------

# Fetch the data
icecreamDF = pd.read_csv(data_processed / 'icecreamsales100random.csv')
print("\nIce Cream Data: ")
print(icecreamDF)


# Totals for All Transactions
print("Totals for All Transactions:")
totalScoops = icecreamDF['Scoops'].sum()
averageScoopSize = icecreamDF['Scoops'].mean()
print("Total Scoops (All Transactions):", totalScoops)
print("Average Scoop Size (All Transactions):", round(averageScoopSize, 2))

# Conditional Subsetting with Row Values
print("Conditional Subsetting: Flavor")
```

```python
# Chocolate
chocolateDF = icecreamDF[icecreamDF['Flavor'] == 'Chocolate']
print("Chocolate Transactions:", chocolateDF.shape[0])
print("Chocolate Total Scoops:", chocolateDF['Scoops'].sum())
print("Chocolate Average Scoops:", round(chocolateDF['Scoops'].mean(), 2))

# Vanilla
vanillaDF = icecreamDF[icecreamDF['Flavor'] == 'Vanilla']
print("Vanilla Transactions:", vanillaDF.shape[0])
print("Vanilla Total Scoops:", vanillaDF['Scoops'].sum())
print("Vanilla Average Scoops:", round(vanillaDF['Scoops'].mean(), 2))

# Blue Moon
blueMoonDF = icecreamDF[icecreamDF['Flavor'] == 'Blue Moon']
print("Blue Moon Transactions:", blueMoonDF.shape[0])
print("Blue Moon Total Scoops:", blueMoonDF['Scoops'].sum())
print("Blue Moon Average Scoops:", round(blueMoonDF['Scoops'].mean(), 2))


print("Conditional Subsetting: Container")

# Waffle Cone
waffleDF = icecreamDF[icecreamDF['Container'] == 'WaffleCone']
print("Waffle Cone Transactions:", waffleDF.shape[0])
print("Waffle Cone Total Scoops:", waffleDF['Scoops'].sum())
print("Waffle Cone Average Scoops:", round(waffleDF['Scoops'].mean(), 2))

# Sugar Cone
sugarDF = icecreamDF[icecreamDF['Container'] == 'SugarCone']
print("Sugar Cone Transactions:", sugarDF.shape[0])
print("Sugar Cone Total Scoops:", sugarDF['Scoops'].sum())
print("Sugar Cone Average Scoops:", round(sugarDF['Scoops'].mean(), 2))

# Groupings
print("Groupings by Flavor:")
print("Transactions by Flavor:")
print(icecreamDF.groupby('Flavor').size())

print("Total Scoops by Flavor:")
print(icecreamDF.groupby('Flavor')['Scoops'].sum())

print("Average Scoops by Flavor:")
print(icecreamDF.groupby('Flavor')['Scoops'].mean().round(2))


print("Groupings by Container:")
print("Transactions by Container:")
print(icecreamDF.groupby('Container').size())

print("Total Scoops by Container:")
print(icecreamDF.groupby('Container')['Scoops'].sum())

print("Average Scoops by Container:")
print(icecreamDF.groupby('Container')['Scoops'].mean().round(2))
```

Shell

```
>>> %Run agg_rand_icdata.py
Alejandro De La Torre's code....
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S 440/Lectures/Week 14/agg_icecream_exercise/code

Ice Cream Data:
   Transaction      Flavor   Container  Scoops
0            1  Strawberry         Cup       3
1            2  Strawberry         Cup       1
2            3  Strawberry   SugarCone       3
3            4     Vanilla  WaffleCone       2
4            5  Strawberry   SugarCone       3
..         ...         ...         ...     ...
95          96   Blue Moon  WaffleCone       3
96          97     Vanilla         Cup       1
97          98  Strawberry   SugarCone       2
98          99  Strawberry         Cup       2
99         100   Chocolate         Cup       1

[100 rows x 4 columns]
Totals for All Transactions:
Total Scoops (All Transactions): 194
Average Scoop Size (All Transactions): 1.94
Conditional Subsetting: Flavor
Chocolate Transactions: 28
Chocolate Total Scoops: 55
Chocolate Average Scoops: 1.96
Vanilla Transactions: 31
Vanilla Total Scoops: 61
Vanilla Average Scoops: 1.97
Blue Moon Transactions: 16
Blue Moon Total Scoops: 31
Blue Moon Average Scoops: 1.94
Conditional Subsetting: Container
Waffle Cone Transactions: 30
Waffle Cone Total Scoops: 59
Waffle Cone Average Scoops: 1.97
Sugar Cone Transactions: 34
Sugar Cone Total Scoops: 71
Sugar Cone Average Scoops: 2.09
Groupings by Flavor:
Transactions by Flavor:
Flavor
Blue Moon     16
Chocolate     28
Strawberry    25
Vanilla       31
dtype: int64
Total Scoops by Flavor:
Flavor
Blue Moon     31
```

```
Chocolate     55
Strawberry    47
Vanilla       61
Name: Scoops, dtype: int64
Average Scoops by Flavor:
Flavor
Blue Moon     1.94
Chocolate     1.96
Strawberry    1.88
Vanilla       1.97
Name: Scoops, dtype: float64
Groupings by Container:
Transactions by Container:
Container
Cup           36
SugarCone     34
WaffleCone    30
dtype: int64
Total Scoops by Container:
Container
Cup           64
SugarCone     71
WaffleCone    59
Name: Scoops, dtype: int64
Average Scoops by Container:
Container
Cup           1.78
SugarCone     2.09
WaffleCone    1.97
Name: Scoops, dtype: float64
>>>
```

```
Transaction,Flavor,Container,Scoops
1,Strawberry,Cup,3
2,Strawberry,Cup,1
3,Strawberry,SugarCone,3
4,Vanilla,WaffleCone,2
5,Strawberry,SugarCone,3
6,Vanilla,SugarCone,3
7,Blue Moon,WaffleCone,1
8,Vanilla,Cup,3
9,Blue Moon,SugarCone,3
10,Blue Moon,Cup,1
11,Blue Moon,Cup,3
12,Chocolate,SugarCone,3
13,Strawberry,Cup,3
14,Vanilla,WaffleCone,3
15,Vanilla,SugarCone,3
16,Strawberry,SugarCone,2
17,Chocolate,WaffleCone,1
18,Chocolate,Cup,1
19,Vanilla,WaffleCone,1
20,Chocolate,SugarCone,3
21,Vanilla,WaffleCone,2
22,Vanilla,SugarCone,2
23,Strawberry,WaffleCone,1
24,Vanilla,Cup,1
```

```
25,Chocolate,SugarCone,1
26,Blue Moon,SugarCone,1
27,Strawberry,Cup,1
28,Strawberry,Cup,3
29,Blue Moon,WaffleCone,3
30,Blue Moon,SugarCone,2
31,Vanilla,SugarCone,2
32,Chocolate,Cup,1
33,Strawberry,WaffleCone,1
34,Blue Moon,SugarCone,3
35,Strawberry,SugarCone,2
36,Strawberry,SugarCone,1
37,Strawberry,Cup,1
38,Chocolate,Cup,1
39,Vanilla,Cup,1
40,Chocolate,WaffleCone,3
41,Chocolate,Cup,1
42,Vanilla,Cup,2
43,Strawberry,Cup,2
44,Vanilla,Cup,3
45,Vanilla,WaffleCone,2
46,Vanilla,Cup,1
47,Vanilla,SugarCone,2
48,Vanilla,SugarCone,3
49,Strawberry,SugarCone,1
50,Chocolate,WaffleCone,1
51,Vanilla,WaffleCone,2
52,Vanilla,Cup,1
53,Vanilla,SugarCone,2
54,Chocolate,WaffleCone,2
55,Chocolate,SugarCone,1
56,Blue Moon,Cup,1
57,Chocolate,SugarCone,3
58,Vanilla,WaffleCone,1
59,Blue Moon,Cup,1
60,Strawberry,Cup,2
61,Chocolate,WaffleCone,1
62,Chocolate,SugarCone,2
63,Chocolate,WaffleCone,3
64,Chocolate,WaffleCone,3
65,Blue Moon,Cup,2
66,Chocolate,SugarCone,3
67,Vanilla,SugarCone,3
68,Strawberry,Cup,3
69,Strawberry,SugarCone,2
70,Chocolate,WaffleCone,2
71,Vanilla,WaffleCone,2
72,Chocolate,SugarCone,3
73,Vanilla,Cup,1
74,Chocolate,WaffleCone,3
75,Chocolate,Cup,2
76,Vanilla,WaffleCone,2
77,Chocolate,WaffleCone,3
78,Vanilla,WaffleCone,1
79,Strawberry,SugarCone,1
80,Blue Moon,Cup,3
81,Strawberry,WaffleCone,2
82,Blue Moon,SugarCone,1
83,Blue Moon,SugarCone,1
84,Strawberry,SugarCone,1
```

```
85,Chocolate,WaffleCone,2
86,Strawberry,WaffleCone,2
87,Chocolate,SugarCone,2
88,Vanilla,Cup,2
89,Strawberry,Cup,2
90,Chocolate,SugarCone,1
91,Vanilla,Cup,2
92,Vanilla,Cup,3
93,Vanilla,WaffleCone,2
94,Blue Moon,Cup,2
95,Chocolate,WaffleCone,2
96,Blue Moon,WaffleCone,3
97,Vanilla,Cup,1
98,Strawberry,SugarCone,2
99,Strawberry,Cup,2
100,Chocolate,Cup,1
```